Biology Final — Quiz Bank

Test: Friday, May 29, 2026

This bank has 60+ practice questions covering all 7 topics. Format mix matches what the review guide suggests: multiple choice, short answer, diagram-style, and "show your work" Punnett squares.

How to use: 1. Cover the answers (they're at the end of each topic). 2. Write your answers on a separate sheet. 3. Grade yourself. 4. For anything wrong: re-read that part of the study guide BEFORE moving on.

TOPIC I — Foundational Science (Sci. Method)

1. A student tests whether plants grow taller with classical music vs. silence. The plant heights are the: A. Independent variable B. Dependent variable C. Control D. Constant

2. In the same experiment, what should be held constant? (Short answer)

3. A scientist measures the percentage of students who prefer pizza vs. pasta vs. salad for lunch. Which graph type is best? A. Line graph B. Bar graph C. Pie chart D. Scatter plot

4. Write a hypothesis for this question: "Does fertilizer affect plant growth?" Use an if/then format.

5. What is the difference between a control group and a constant?

Answers — Topic I

1. **B** (heights are what you measure; they depend on the treatment)
2. Anything that could affect plant growth other than the music: water amount, sunlight, soil type, pot size, temperature, plant species.
3. **C** (parts of a whole — percentages adding to 100%)
4. "If plants are given fertilizer, then they will grow taller than plants without fertilizer." (Any if/then with a measurable outcome works.)
5. **Control group** = the comparison group that gets no treatment. **Constants** = variables kept the same across ALL groups so they don't affect the result.

TOPIC II — RNA & Protein Synthesis

6. Write the central dogma using arrows.

7. Where does transcription occur? Where does translation occur?

8. Name 3 ways RNA differs from DNA.

9. What's the difference between a codon and an anticodon?

10. Transcribe this DNA into mRNA: `TAC-GGA-TTC-ACG-ATT`

11. Using the mRNA from #10 and a codon wheel, what amino acid sequence does it make? (You'll need the codon wheel; the first codon is the start codon.)

12. Identify each mutation: a. Original: AUG-CCA-GUA → Mutated: AUG-CCG-GUA b. Original: AUG-CCA-GUA → Mutated: AUG-CCA-GUA-AUG (an extra base added at position 4) c. Original: AUG-CCA-GUA → Mutated: AUG-UAA-GUA

13. Which type of mutation is most likely to produce a non-functional protein: a silent substitution, a missense substitution, or a frameshift? Why?

14. What is differentiation? Why does it matter that the genetic code is the same in every cell but different cells make different proteins?

Answers — Topic II

1. **DNA → (transcription) → RNA → (translation) → Protein**
2. **Transcription = nucleus** (where DNA lives). **Translation = cytoplasm** (at the ribosome).
3. (any 3) RNA is single-stranded / DNA is double-stranded · RNA has ribose / DNA has deoxyribose · RNA uses uracil (U) / DNA uses thymine (T) · RNA can leave the nucleus / DNA cannot.
4. **Codon** is on the mRNA (3 bases that code for an amino acid). **Anticodon** is on the tRNA (3 bases that base-pair with the codon to bring the right amino acid).
5. mRNA: `AUG-CCU-AAG-UGC-UAA`. (Remember: A→U, T→A, G→C, C→G. Wherever DNA had T, mRNA has A; wherever DNA had A, mRNA has U.)
6. AUG = Met (start) · CCU = Pro · AAG = Lys · UGC = Cys · UAA = STOP. Sequence: **Met-Pro-Lys-Cys-STOP**.
7. a. **Silent substitution** (CCA and CCG both code for Pro — no amino acid change). b. **Insertion → frameshift** (every codon downstream is now misread). c. **Nonsense substitution** (UAA is a stop codon — protein gets cut short).
8. **Frameshift** — every single codon downstream of the insertion/deletion is changed, so the rest of the protein is essentially gibberish. A silent substitution does nothing; a missense changes 1 amino acid (often tolerable).
9. **Differentiation** = the process where cells become specialized (skin, nerve, muscle, etc.). Every cell has the SAME DNA, but different genes are TURNED ON in different cells. That's why a liver cell looks and acts nothing like a neuron, even though they're genetically identical.

TOPIC III — Meiosis

15. Why must the chromosome number be HALVED during meiosis?

16. Why does meiosis in a female produce only ONE egg (instead of 4 like sperm)?

17. A human body cell has 46 chromosomes. After meiosis, how many chromosomes does each gamete have?

18. Name 2 events during meiosis that create genetic variation.

19. Give 3 differences between mitosis and meiosis.

20. Put these meiosis phases in order: Telophase II, Prophase I, Metaphase I, Anaphase II, Prophase II, Anaphase I, Telophase I, Metaphase II.

21. During which phase does crossing over occur?

Answers — Topic III

1. So that when sperm (n) + egg (n) join at fertilization, the zygote ends up with the correct DIPLOID number (2n) — not double it. Without halving, every generation would double the chromosome count.
2. Because the egg needs to keep ALL the cytoplasm and nutrients to support the developing embryo. The other 3 cells become polar bodies (small, no cytoplasm) and disintegrate.
3. **23** (haploid).
4. **Crossing over** (Prophase I — homologous chromosomes swap segments) AND **independent assortment** (Metaphase I — homologous pairs line up randomly, so each gamete gets a random mix of mom's/dad's chromosomes).
5. Mitosis → 2 identical diploid cells; Meiosis → 4 non-identical haploid cells. · Mitosis = 1 round of division; Meiosis = 2 rounds. · Mitosis = asexual reproduction / body cells; Meiosis = sexual reproduction / gametes. · No crossing over in mitosis; crossing over happens in meiosis.
6. Prophase I → Metaphase I → Anaphase I → Telophase I → Prophase II → Metaphase II → Anaphase II → Telophase II.
7. **Prophase I**.

TOPIC IV — Genetics

22. State Mendel's three laws (Dominance, Segregation, Independent Assortment) in one sentence each.

23. In pea plants, Yellow (Y) is dominant over green (y). Cross a heterozygous yellow (Yy) with a green (yy). Show the Punnett square. What's the genotype ratio? Phenotype ratio?

24. A dihybrid cross: TtYy × TtYy (T = tall dominant, t = short; Y = yellow dominant, y = green). Predict the phenotype ratio. (Hint: should be 9:3:3:1 for a standard dihybrid heterozygous cross.)

25. Snapdragons show **incomplete dominance**: Red (RR) × White (WW) gives all PINK (RW). If you cross two pink snapdragons (RW × RW), what's the offspring ratio?

26. A human with blood type AB marries someone with blood type O. What blood types are possible in their children?

27. Color-blindness is X-linked recessive. A carrier mother (X^Bx^b) and a normal father (X^BY) have a child. What's the probability the child is a color-blind son?

28. What does a karyotype show? Can you tell biological sex from one? How?

29. A pedigree shows a trait that appears in every generation, in both males and females, and an affected parent always has at least one affected child. Is the trait dominant or recessive? Autosomal or sex-linked?

30. Use a Punnett square to show how a person with blood type A (genotype I^Ai) and a person with blood type B (genotype I^Bi) could have a child with any of the 4 blood types. Show your work.

Answers — Topic IV

22. - **Law of Dominance:** One allele can mask (be dominant over) another. - **Law of Segregation:** During gamete formation, the two alleles for each gene separate so each gamete has only one. - **Law of Independent Assortment:** Alleles for different genes separate INDEPENDENTLY of each other. 23. Punnett square: `Yy × yy` → offspring are Yy, Yy, yy, yy. **Genotype ratio: 2 Yy : 2 yy (or 1:1). Phenotype ratio: 2 yellow : 2 green (1:1, or 50% yellow / 50% green).** 24. **9 tall yellow : 3 tall green : 3 short yellow : 1 short green** (classic 9:3:3:1). 25. **1 RR (red) : 2 RW (pink) : 1 WW (white)** — both genotype and phenotype ratios are 1:2:1 because incomplete dominance shows all 3 phenotypes. 26. AB ($I^A I^B$) × O (ii). Punnett: I^Ai, I^Ai, I^Bi, I^Bi. **Children can be either Type A or Type B. NEVER Type AB or Type O.** 27. Punnett: $X^BX^b \times X^BY$. Children: X^BX^B (girl, normal), X^BX^b (girl, carrier), X^BY (boy, normal), X^bY (boy, color-blind). **Probability of color-blind son = 1 in 4 (25%).** (Or, if you only count sons: 1 in 2 of the sons are color-blind.) 28. A karyotype shows all chromosomes lined up by size and pair. Yes — biological sex: **XX = female, XY = male** (look at pair 23). 29. **Autosomal dominant.** Appears every generation, affects both sexes equally, and affected children come from affected parents. 30. $I^Ai \times I^Bi$ → $I^A I^B$ (AB), I^Ai (A), I^Bi (B), ii (O). **Children can be any of the four blood types — all are possible**, each with 25% probability. (This is a great demo of codominance + recessive in one cross.)

TOPIC V — Evolution

31. Summarize Darwin's theory of evolution by natural selection in 2–3 sentences.

32. Name 2 thinkers who influenced Darwin, and what each contributed.

33. List 4 lines of evidence for evolution.

34. Give an example of natural selection actually observed (real-world).

35. What are 3 sources of genetic variation in a population?

36. Describe directional, stabilizing, and disruptive selection in one sentence each.

37. What's the difference between the bottleneck effect and the founder effect?

38. Name 3 types of reproductive isolation that can lead to speciation.

39. What is a cladogram? What do the branches and nodes represent?

40. Two birds that look identical live on the same island but mate at different times of year. Are they the same species? Why or why not?

Answers — Topic V

1. Individuals in a population vary. Some variations help an organism survive and reproduce better in its environment (higher fitness). Those individuals leave more offspring, so the helpful variation becomes more common over generations. Over very long time, this produces new species and explains the diversity AND unity of all life from common ancestors.
2. **Hutton & Lyell** — the Earth is very old, and slow geological processes shape it (uniformitarianism). This gave Darwin enough time for evolution to happen. **Lamarck** — proposed evolution through inheritance of acquired traits (wrong about mechanism, but right that species change). **Malthus** —

populations grow faster than resources, so most offspring don't survive. This is the "struggle for existence" Darwin built on.

3. (any 4) **Biogeography** (where species live now and lived in the past) · **Fossils** (especially transitional fossils like whale ancestors) · **Homologous structures** (same anatomy, different function) · **Vestigial structures** (leftover unused parts) · **Embryology** (similar embryonic development across species) · **Molecular biology** (DNA / proteins like cytochrome C are more similar in closely related species).
4. (any one) Drug-resistant bacteria (e.g., antibiotic resistance). Pesticide-resistant insects. Influenza or COVID-19 evolving new strains. Peppered moths becoming darker during the Industrial Revolution. Galapagos finch beak sizes changing with drought.
5. **Mutations**, **crossing over** (during meiosis), **independent assortment** (during meiosis), **sexual reproduction** (mixing of two parents' DNA at fertilization).
6. **Directional** — one extreme is favored, the bell curve SHIFTS. **Stabilizing** — the middle is favored, the bell curve NARROWS. **Disruptive** — both extremes are favored, the bell curve SPLITS into two peaks.
7. **Bottleneck** = a population gets DRASTICALLY REDUCED (disaster, disease), and the survivors carry only some of the original alleles. **Founder** = a SMALL GROUP leaves the main population and starts a new one — their allele frequencies become the new "normal." Both are types of genetic drift.
8. **Behavioral** (different courtship — meadowlark songs), **Geographic** (physical barrier — squirrels separated by a canyon), **Temporal** (different timing — two flowers blooming at different times).
9. A branching diagram showing evolutionary relationships. **Branches** = lineages. **Nodes** (branch points) = common ancestors. Species closer together on the tree share more recent common ancestors.
10. **They are now separate species** by the biological species definition — they're reproductively isolated by **temporal isolation** (different mating times means they can't interbreed). Even though they LOOK identical, if they can't successfully breed, they're separate species.

TOPIC VI — Ecology

41. List the levels of biological organization from smallest to largest, starting at "species."

42. Give 3 examples of abiotic factors and 3 examples of biotic factors.

43. What is the 10% rule? If a tree captures 10,000 units of energy, roughly how much reaches a hawk that eats a snake that eats a mouse that eats a bug that eats the tree?

44. Identify each relationship as mutualism, commensalism, or parasitism: a. Tapeworm in a dog's intestine. b. Bee pollinating a flower and getting nectar. c. Barnacle attached to a whale's skin — the barnacle gets transportation, the whale isn't affected.

45. What is a keystone species? Give an example.

46. What is the difference between primary succession and secondary succession?

47. A density-INdependent limiting factor would be… (give an example).

48. A density-dependent limiting factor would be… (give an example).

49. Why is energy said to FLOW through an ecosystem while matter CYCLES?

50. What is competitive exclusion? Give an example or explain why two species can't share exactly the same niche.

51. A pond gets too much nitrogen runoff. Algae blooms, then dies, then bacteria decompose it and use up all the oxygen. Fish die. Identify: limiting nutrient, density-dependent vs. independent factors, and which trophic level was disturbed.

Answers — Topic VI

1. Species → Population → Community → Ecosystem → Biome → Biosphere.
2. Abiotic: sunlight, water, soil, temperature, air, minerals (any 3). Biotic: plants, animals, fungi, bacteria, protists (any 3).
3. The 10% rule says only ~10% of energy passes to the next trophic level. So: tree (10,000) → bug (1,000) → mouse (100) → snake (10) → hawk (**~1 unit**). That's why food chains rarely have more than 4–5 levels — there's just not enough energy.
4. a. **Parasitism** (+/-). b. **Mutualism** (+/+). c. **Commensalism** (+/0).
5. A species whose presence has a disproportionately large effect on its community. Classic example: **sea otters** — they eat sea urchins, which keeps urchins from destroying kelp forests. Remove otters → urchins explode → kelp disappears → entire ecosystem collapses.
6. **Primary** starts on bare rock with NO soil (after a volcanic eruption or glacial retreat). Pioneer species like lichens must build soil first. Slow. **Secondary** starts where a community was disturbed but soil REMAINS (after a wildfire, hurricane, or abandoned farm). Faster.
7. **Density-independent** = affects population regardless of size: weather extremes, natural disasters (drought, hurricane, fire, cold snap).
8. **Density-dependent** = stronger as population gets denser: competition, predation, disease, parasitism, herbivory.
9. **Energy is lost as heat at each step** (and you can't reuse heat), so it flows in one direction sun → producers → consumers → heat. **Matter (carbon, water, nitrogen, etc.) is not destroyed** — it's recycled by decomposers back into the environment for new producers to use.
10. **Competitive exclusion principle:** no two species can occupy exactly the same niche in the same habitat at the same time. One will out-compete the other, and the loser will either move, change its niche, or go locally extinct.
11. Limiting nutrient: **nitrogen** (its excess triggered the cascade — and oxygen became the limiting factor for fish). Type of factor: **density-dependent** (the more algae/bacteria, the worse the oxygen depletion). Trophic levels disturbed: primarily producers (algae bloom + death) and consumers (fish killed). This is called eutrophication.

TOPIC VII — Human Body

Reproductive System

52. What's the function of the scrotum being OUTSIDE the body cavity?

53. Trace the path of a sperm from where it's made until it leaves the body. (List the structures in order.)

54. Name the 4 phases of the menstrual cycle in order. On approximately which day does ovulation occur?

55. Match each hormone with its function: a. FSH i. Triggers release of egg from ovary b. LH ii. Stimulates follicle to develop / stimulates spermatogenesis c. Estrogen iii. Stimulates growth of uterine lining; prepares for implantation d. Progesterone iv. Stimulates oogenesis + female secondary sex characteristics e. Testosterone v. Drives sperm production + male secondary sex characteristics

Circulatory System

56. Trace the path of a drop of blood starting at the right atrium. Include: heart chambers, valves, major vessels, and the lungs/body. End back at the right atrium.

57. Label the chamber: a. Receives deoxygenated blood from the body. b. Pumps oxygenated blood to the body. c. Receives oxygenated blood from the lungs. d. Pumps deoxygenated blood to the lungs.

58. Which side of the heart is oxygenated, and which is deoxygenated?

59. What's the difference between an artery, a vein, and a capillary? Include direction of blood flow and one structural feature of each.

60. Pulmonary artery vs. pulmonary vein: which one carries deoxygenated blood? (This is a classic trick question.)

Respiratory System

61. Trace the path of an O_2 molecule from the moment you inhale until it gets into your blood. List every structure in order.

62. What's the function of cilia + mucus in the airway?

63. Is it better to breathe through your nose or your mouth? Why?

64. Explain how the diaphragm makes you inhale.

65. Where does gas exchange actually happen, and why is that location well-suited for it (give 2 features)?

Answers — Topic VII

1. The testes need to be slightly COOLER than core body temperature for sperm production to work properly. The scrotum hangs outside the body to keep them cooler.
2. Testes → epididymis → vas deferens → joins with seminal fluid from the prostate (becoming semen) → urethra (through the penis) → out.
3. **Follicular phase** → **Ovulation (~day 14)** → **Luteal phase** → **Menstruation** (if no fertilization). Then it restarts.
4. a–ii (FSH = stimulates follicle / spermatogenesis), b–i (LH = triggers ovulation), c–iv (Estrogen = oogenesis + female 2° sex characteristics), d–iii (Progesterone = uterine lining), e–v (Testosterone = sperm + male 2° sex characteristics).
5. Right atrium → tricuspid valve → right ventricle → pulmonary semilunar valve → pulmonary artery → **LUNGS** (gas exchange: drops off CO_2, picks up O_2) → pulmonary veins → left atrium → bicuspid (mitral) valve → left ventricle → aortic semilunar valve → aorta → **BODY** (gas exchange in capillaries: drops off O_2, picks up CO_2) → vena cavae (superior + inferior) → right atrium. **Loop complete.**
6. a. Right atrium. b. Left ventricle. c. Left atrium. d. Right ventricle.
7. **Right = deoxygenated** (heading to lungs). **Left = oxygenated** (heading to body).
8. **Artery** — carries blood AWAY from heart, thick muscular walls, high pressure (where you feel your pulse). **Vein** — carries blood TO heart, thinner walls, contains valves to prevent backflow. **Capillary** — connects arteries to veins, one cell thick, site of exchange with tissues.
9. **Pulmonary artery carries DEOXYGENATED blood** (right ventricle → lungs). This is the ONLY artery that carries deoxygenated blood. The pulmonary VEIN, in contrast, carries oxygenated blood — the

only vein that does so. Remember: "artery = away from heart" and "vein = into heart," regardless of oxygen content.

10. Nasal cavity (or mouth) → pharynx → larynx → trachea → bronchi → bronchioles → alveoli → diffuses across alveolar wall → capillary → red blood cell (binds to hemoglobin).
11. **Mucus traps** dust, pollen, microbes, and other particles. **Cilia sweep** the mucus + trapped debris UP and OUT of the airway (toward the throat where you swallow or cough it out). They protect the lungs from infection and damage.
12. **Through your NOSE.** The nasal cavity warms, moistens, and filters the air (mucus + cilia + hairs). Mouth-breathing skips all that filtration and lets in cold, dry, unfiltered air — harder on the lungs.
13. The diaphragm CONTRACTS and flattens DOWNWARD, which expands the chest cavity → drops air pressure inside the lungs below atmospheric pressure → air rushes IN. (To exhale: diaphragm relaxes, chest gets smaller, pressure inside lungs rises, air pushes out.)
14. Gas exchange happens in the **alveoli** (with the capillaries that surround them). Good for gas exchange because: (1) **HUGE surface area** — millions of tiny alveoli pack a tennis-court's worth of surface into your chest, (2) **walls are extremely thin** (one cell thick) so oxygen and CO_2 diffuse across quickly, (3) **moist** environment helps gases dissolve, (4) **dense capillary network** keeps blood constantly available to pick up O_2 and drop off CO_2.

Bonus: Cross-topic synthesis

66. Connect meiosis to evolution: Why does meiosis matter for evolution? **Answer:** Meiosis creates the genetic variation that natural selection acts on. Without crossing over + independent assortment + sexual reproduction (each meiosis produces a unique gamete), every offspring would be genetically identical and there'd be nothing for natural selection to "select" between. Mutation provides new alleles; meiosis shuffles them into new combinations every generation.

67. Connect the circulatory and respiratory systems: How do they work together? **Answer:** The respiratory system gets O_2 in and CO_2 out of the body, but only at the alveoli. The circulatory system then carries that O_2 (via red blood cells) from the lungs to every cell, and brings back the CO_2 waste from those cells to the lungs to be exhaled. They can't function without each other — that's why "cardio-respiratory" is one phrase.

68. Connect genetics + evolution: What is the relationship between alleles, gene pools, and evolution? **Answer:** Genes have multiple alleles. The gene pool is all the alleles in a population. Evolution IS, in genetic terms, a change in allele frequencies in the gene pool over time. Natural selection acts on phenotypes, but the result shows up as shifting allele frequencies — that's how evolution and Mendelian genetics fit together.